13-4b Economies and Diseconomies of Scale

The shape of the long-run average-total-cost curve conveys important information about a firm's production processes. In particular, it tells us how costs vary with the scale—that is, the size—of a firm's operations. When long-run average total cost declines as output increases, there are said to be **economies of scale** (the property whereby long-run average total cost falls as the quantity of output increases). When long-run average total cost rises as output increases, there are said to be **diseconomies of scale** (the property whereby long-run average total cost rises as the quantity of output increases). When long-run average total cost does not vary with the level of output, there are said to be **constant returns to scale** (the property whereby long-run average total cost stays the same as the quantity of output changes). In Figure 6, Ford has economies of scale at low levels of output, constant returns to scale at intermediate levels of output, and diseconomies of scale at high levels of output.

What might cause economies or diseconomies of scale? Economies of scale often arise because higher production levels allow *specialization* among workers, which permits each worker to become better at a specific task. For instance, if Ford hires a large number of workers and produces a large number of cars, it can reduce costs using modern assembly-line production. Diseconomies of scale can arise because of *coordination problems* that often occur in large organizations. The more cars Ford produces, the more stretched the management team becomes, and the less effective the managers become at keeping costs down.

This analysis shows why long-run average-total-cost curves are often U-shaped. At low levels of production, the firm benefits from increased size because it can take advantage of greater specialization. Coordination problems, meanwhile, are not yet acute. By contrast, at high levels of production, the benefits of specialization have already been realized, and coordination problems become more severe as the firm grows larger. Thus, long-run average total cost is falling at low levels of production because of increasing specialization and rising at high levels of production because of growing coordination problems.

FYI

Lessons from a Pin Factory

"Jack of all trades, master of none." This old adage sheds light on the nature of cost curves. A person who tries to do everything usually ends up doing nothing very well. If a firm wants its workers to be as productive as they can be, it is often best to give each worker a limited task that she can master. But this organization of work is possible only if a firm employs many workers and produces a large quantity of output.

In his book *The Wealth of Nations*, Adam Smith described a visit he made to a pin factory. Smith was impressed by the specialization among the workers and the resulting economies of scale. He wrote,

> *One man draws out the wire, another straightens it, a third cuts it, a fourth points it, a fifth grinds it at the top for receiving the head; to make the head requires two or three distinct operations; to put it on is a peculiar business; to whiten it is another; it is even a trade by itself to put them into paper.*

Smith reported that because of this specialization, the pin factory produced thousands of pins per worker every day. He conjectured that if the workers had chosen to work separately, rather than as a team of specialists, "they certainly could not each of them make twenty, perhaps not one pin a day." In other words, because of specialization, a large pin factory could achieve higher output per worker and lower average cost per pin than a small pin factory.

The specialization that Smith observed in the pin factory is common in the modern economy. If you want to build a house, for instance, you could try to do all the work yourself. But you would more likely turn to a builder, who in turn hires carpenters, plumbers, electricians, painters, and many other types of workers. These workers focus their training and experience in particular jobs, and as a result, they become better at their jobs than if they were generalists. Indeed, the use of specialization to achieve economies of scale is one reason modern societies are as prosperous as they are.

QuickQuiz

8. If a higher level of production allows workers to specialize in particular tasks, a firm will likely exhibit ____ of scale and ____ average total cost.

    a. economies; falling

    b. economies; rising

    c. diseconomies; falling

    d. diseconomies; rising

9. If Boeing produces 9 jets per month, its long-run total cost is \$9 million per month. If it produces 10 jets per month, its long-run total cost is \$11 million per month. Boeing exhibits

   a. rising marginal cost.

   b. falling marginal cost.

   c. economies of scale.

   d. diseconomies of scale.

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